

A Topological Subcase of the Union Formula for (I)-Envelopes

Candidate partial result for arXiv:2303.07691

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Source and Open Question

The source paper is Ondrej F. K. Kalenda and Matias Raja, *Topologies related to (I)-envelopes*, arXiv:2303.07691 [1].

Question 5.1 of the source paper asks whether, for a Banach space X and convex sets $A, B \subset X^*$ such that $A \cup B$ is also convex, one always has

$$(I)\text{-env}(A \cup B) = (I)\text{-env}(A) \cup (I)\text{-env}(B).$$

The question is motivated by the topological-closure behavior needed for the non-locally-convex version of the paper's Question 1.1.

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5. FINAL REMARKS AND OPEN PROBLEMS

The answer to the ‘locally convex’ variant of Question 1.1 is for some Banach spaces positive and for some Banach spaces negative as witnessed by the results of Section 3. However, a characterization of those Banach spaces for which the answer is positive is still missing as pointed in Question 3.9.

The ‘topological’ variant of Question 1.1 is widely open. We have no counterexample to it, but we do not have either any example for which the ‘topological’ variant has a positive answer but the ‘locally convex’ variant has a negative answer. In view of Theorem 4.3 the following problem is natural to ask:

Question 5.1. *Assume that X is a Banach space and $A, B \subset X^*$ are two convex sets such that $A \cup B$ is also convex. Is $(I)\text{-env}(A \cup B) = (I)\text{-env}(A) \cup (I)\text{-env}(B)$?*

Further, Question 3.10 suggests that there may be a substantial difference between (I)-envelopes of bounded and unbounded sets. It seems that the following question is interesting and open.

Proof Intuition

The general question is difficult because it asks whether the operation $A \mapsto (I)\text{-env}(A)$ has a finite-union behavior that ordinary topological closures have automatically. If, however, the (I)-envelope is already known to come from closure in a fixed topology after taking convex hulls, the desired identity is immediate. The hypotheses in Question 5.1 make A , B , and $A \cup B$ convex, so the convex-hull operation does not add anything; all that remains is the elementary identity

$$\overline{A \cup B}^\tau = \overline{A}^\tau \cup \overline{B}^\tau.$$

The source theorem supplies such a topology when X contains no copy of ℓ^1 .

Theorem

Theorem 1. *Let X be a Banach space. Suppose that there is a topology τ on X^* such that*

$$(\text{I-env})(S) = \overline{\text{co } S}^\tau$$

for every $S \subset X^$. Then for every pair of convex sets $A, B \subset X^*$ such that $A \cup B$ is convex,*

$$(\text{I-env})(A \cup B) = (\text{I-env})(A) \cup (\text{I-env})(B).$$

Consequently, the equality holds for every Banach space X containing no isomorphic copy of ℓ^1 .

Proof. Let $C = A \cup B$. Since A , B , and C are convex, we have

$$\text{co } A = A, \quad \text{co } B = B, \quad \text{co } C = C.$$

Using the assumed topological representation of the (I)-envelope and the finite-union law for closure in any topological space, we obtain

$$(\text{I-env})(A \cup B) = \overline{C}^\tau = \overline{A \cup B}^\tau = \overline{A}^\tau \cup \overline{B}^\tau = (\text{I-env})(A) \cup (\text{I-env})(B).$$

For the final assertion, apply Theorem 3.3 of [1]: if X contains no isomorphic copy of ℓ^1 , then for every $S \subset X^*$,

$$(\text{I-env})(S) = \overline{\text{co } S}^{\sigma(X^*, B_1(X))}.$$

Thus the preceding argument applies with $\tau = \sigma(X^*, B_1(X))$. □

Verification Notes

The proof uses only two inputs.

- The closure of a finite union is the union of the closures in every topological space.
- The source paper's Theorem 3.3 represents all (I)-envelopes by the locally convex topology $\sigma(X^*, B_1(X))$ when X contains no copy of ℓ^1 .

The convexity assumption in Question 5.1 is essential for this short proof: without it, the terms would be closures of $\text{co}(A \cup B)$, $\text{co } A$, and $\text{co } B$, and the finite-union closure identity would not directly apply.

Limitations and Novelty Check

This is only a partial result. It does not answer Question 5.1 for spaces where no topology is known to represent all (I)-envelopes, and it gives no new counterexample to the topological version of Question 1.1.

The active run indexes were searched for arXiv:2303.07691, the source title, “(I)-envelope”, “Ienv(A union B)”, “Ienv(A) union Ienv(B)”, and related topology keywords. A bounded web search on June 14, 2026 for exact union-formula phrases found only the source paper and no later explicit answer. Since this packet is a source-theorem corollary and only a subcase, it should be reviewed as a modest partial result.

Human Review Recommendation

Check that the cited source theorem applies to arbitrary subsets $S \subset X^*$, not merely to convex or bounded sets. Then verify that the convexity assumptions in Question 5.1 are exactly those used to eliminate the convex hulls.

References

- [1] O. F. K. Kalenda and M. Raja, *Topologies related to (I) -envelopes*, arXiv:2303.07691v2, 2023.